SYS

TIM2

stm32F746

global interrupt

Software Architecture

interrupt based process at 40us

Communication Profile

Command words
Status words
References

CAN-BUS

TX/RX

Global State Machine

Global control
Automatic Command Word Gen.
Faults management
Profile communication

Inverter State Machine

Inverter state machine
Peripherals management
Faults management

Peripherals

ADCs Acquisitions
GPIOs

Peripherals

PWM-outputs management
MOSFET Drivers management
OCDs (as interrupts)

Inverter Control

Mode of operation
Faults management
DQ-current transformation

Mode of operation

Speed control
PMSM vector current control
I/F current control
U/F open loop control
Signal phase generator

BEMF based model Observer

Rotor Phase estimation based
on BEMF model observer